

Exhibit 3: Thinning Inspection Plot Form and Detailed Instructions

THINNING INSPECTION PLOT FORM

sheet ____ of ____

ID	(a) Forest	(b) District	(c) Stand #	(d) Inspector (Gov't)	(e) Date		
Contract	(f) Contractor	(g) Sub-item #	(h) Acres	(i) % of Work Accomplished			
	(j) Contract #	(k) Contract Inspector		(l) % of Contract Time Left			
Contract Specifications	(m) Acceptable tpa ____min/ ____max ____avg.	(n) Accept. Trees/plot ____min/ ____max ____avg.	(o) Avg Spacing X ____ foot spacing	(p) Plot Size 1: ____ or 1/ ____ acre	(q) Min. Cut Ht ____in.	(r) Max. cut dbh ____in.	(s) Max. Cat C ____tpa

Plot No.	Code Species	Tree Evaluation and Species	SATISFACTORY			DEFICIENCY				
			Satisfactory Leave Trees	Surplus Trees	Non-stocked points	Cat. "A" trees Improper Leave (Uncorrectable)	Cat. "B" trees (Correctable)	Cat. "C" (Correctable)		
(1)	Species ? Tree was cut	Codes ?	S1-S2	S3-S4	N1-N2	A1-A2	A3-A4	B1-B2	B3-B4	C1-C6
	1 2 3 4 5 6 7 8 9 10 11 12	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Remarks:			Avg D.B.H. = ____"			Avg HT = ____'		Cat. C tally:		

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Remarks:			Avg D.B.H. = ____"			Avg HT = ____'		Cat. C tally:		
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(t) Thinning Quality Calculation	Totals of this page								
	Total of all pages								
	(3) Accept. Leave	(4) Surplus	(5) Non-stocked	(6) Cat A	(7) Cat A	(8) Cat B	(9) Cat B	(10) Cat C	

Backside

INSTRUCTIONS

1. Locate and mark plot center on the ground.
2. Locate and examine the trees selected for leave, and trees that are cut on the plot. Record the Evaluation Code for satisfactory trees and deficiencies (except Category C trees) in column 2 on the top line for the plot. If required, record the species for each tree in the second line for the plot.
3. Tally all Category C deficiencies and record in the space below Column 10. (Tally by code- i.e. C1, C2 C3)
4. Total the Acceptable Leave Trees coded S1, S2 in column 3. Total cannot exceed the max. trees per plot. If total exceeds the max/plot, re-code trees [i.e. surplus trees over max. DBH should be coded S3, trees to ignore should be coded S4. Excess Trees are a deficiency, Category B]. Refer to Block n for maximum trees per plot.
5. Total the surplus trees >max. d.b.h (S3) and trees to ignore (S4) and record in column 4.
6. Total the non-stocked points on the plot coded N1 and N2 and record in column 5.
7. Total the improper selection and cutting of trees (A1, A2) that should have been left; record in column 6.
8. Total the improper selection of leave trees (A3) or damage to leave trees (A4), record in column 7.
9. Total the trees that should have been cut (B1, B2), record in column 8.
10. If required, total the trees that should have been girdled or pruned or that were girdled or pruned incorrectly, record in column 9.
11. Total the Category C trees on the plot and record in column 10.
12. If required: estimate the average d.b.h and average height of the leave trees in the plot.
13. Remarks: include other required information
14. Total Columns 3-10 on each page and provide the grand total of all pages on page 1.
15. After all plots have been conducted, calculate thinning quality using the total of all pages. Record in block t.

$$\% \text{ Quality} = 1.00 - \left(\frac{\text{Cat A deficiencies (col. 6 + col. 7)} + \text{Cat B deficiencies (col. 8 + col. 9)}}{\text{Acceptable Leave (col. 3) + Non-stocked points (col. 5)}} \right) \times 100$$

16. Determine if Cat C trees (col. 10) exceed the maximum allowed specified (block s).

$$\text{Cat C tpa} = \frac{\text{Total Col 10}}{\text{No. of plot}} \times \text{reciprocal of plot size}$$

TREE EVALUATION CODES

Satisfactory Leave Trees

- S1 (or ?) Leave Tree meets specifications
- S2 Tree left exceeds maximum d.b.h to cut, included in trees/plot calculation
- S3 Tree left exceeds maximum d.b.h to cut, not included in trees/plot calculation
- S4 Tree to be ignored, not included in trees/plot calculation.

Non-stocked areas

- N1 Opening, no tree exists
- N2 Opening created by cutting a tree with insect/disease or poor form.

Improperly cut trees: (Category A) Uncorrectable

- A1 Tree should not have been cut, tree met leave tree specifications.
- A2 Tree should not have been cut, tree exceeds max. d.b.h to cut
- A3 Leave tree damaged by thinning operations
- A4 Improper selection of leave tree (includes wrong species, undesirable tree, etc.)

Excess Trees: (Category B) Correctable

- B1 Tree should have been cut to meet allowable stocking density or is too close to leave tree.
- B2 Tree should have been cut because of damage, insect & disease or poor form.
- *B3 Tree should have been girdled or was girdled improperly (or Cat C5)
- *B4 Tree should have been pruned or was pruned improperly (or Cat C6)

Other Deficiencies: (Category C) Correctable

- | | |
|---|---|
| C1 Tree was not completely severed from stump | C4 Small tree |
| C2 Hang up tree | *C5 Tree should have been girdled or was girdled improperly (or Cat B3) |
| C3 High Stump | *C6 Tree should have been pruned or was pruned improperly (or Cat B4) |

Additional Notes (local requirements, etc.)

** If girdling or pruning is required, specify if trees not girdled/pruned or girdled/pruned improperly are considered a Cat B or Cat C deficiency prior to start work. Define a proportional value for each pruning deficiency if desired (i.e. 0.25 per deficiency)*

Exhibit 4: Inspection Plot form and Detailed Instructions, (Continued)

DETAILED INSTRUCTIONS TREE EVALUATION CODES

	Codes	Explanation
SATISFACTOR	SATISFACTORY LEAVE TREES The totals of S1 and S2 trees are used in the thinning/slashing quality calculation.	
	S1 or check mark	Acceptable tree: tree was left that met the leave tree specifications.
	S2	Tree was left that exceeds the maximum d.b.h. to cut, and the prescription/contract specified this tree counts in the acceptable trees per plot calculation when no tree within the thinning range exists.
	S3	Tree was left that exceeds the maximum d.b.h. to cut but is considered a surplus tree and is not included in the acceptable trees per plot. This value is important to record in order to calculate total trees per acre in the post treated stands. Refer to the prescription, which will specify if these larger trees are considered surplus in all cases, or only when there are sufficient trees within the thinning range to meet the allowable density.
	S4	Tree was left that was of the species or condition that was to be ignored; it is considered a surplus tree and not included in the acceptable trees per plot. This value is important to record as it is needed to calculate total trees per acre in the post treated stands.
	NON-STOCKED AREAS The totals of non-stocked areas are used in the thinning quality calculation.	
	N1	An opening exists on the plot where no tree exists. Recording this on the plot form justifies fewer than minimum trees on a plot.
	N2	An opening that was created by cutting a tree due to poor form, insect and disease or other conditions as required by the contract or prescription. Recording this gives credit for cutting a tree that should be cut even if it lowers the stocking below minimum levels.
DEFICIENT	IMPROPERLY CUT TREES – CATEGORY A DEFICIENCIES These deficiencies are uncorrectable and influence the thinning/slashing quality calculation. Regardless of the thinning/slashing quality, rework cannot be done to correct these problems.	
	A1	Tree was cut that should not have been it that met all leave tree specifications, and did not exceed the maximum trees per plot.
	A2	Tree was cut that should not have been because it exceeded maximum d.b.h. to cut. This could instead be coded A1 but is more specific using this code.
	A3	Satisfactory leave tree was damaged in the thinning operations; may have been damaged during cutting, slash treatment or any aspect of the operations.
	A4	Wrong leave tree was left. It was the wrong species; a better tree existed in the spacing or competitive area, or for similar reasons. Do not also code the tree that was incorrectly cut as a deficiency

DEFICIENCIES	<u>EXCESS TREES REMAIN – CATEGORY B DEFICIENCIES</u> These deficiencies are correctable and influence the thinning/slashing quality calculation. Reworking the area and cutting more trees can correct these errors.	
	B1	Tree met tree specifications but should have been cut to meet the allowable stocking density, or it was too close to a leave tree.
	B2	Tree should have been cut because of damage, insect and disease or poor form or otherwise indicated as a tree that is to be cut regardless of other trees on the plot. Refer to prescription for unacceptable levels of insect, disease or damage or the characteristics of other trees to be cut.
	B3	When girdling is required, this code indicates that a tree was not girdled or not adequately girdled that should have been. The prescription/contract must specify if this will be considered a B or C deficiency. As a B deficiency, there is less tolerance and will affect the quality calculation.
	B4	When pruning is required, this code indicates that a tree was not pruned or not adequately pruned that should have been. The prescription/contract must specify if this will be considered a B or C deficiency. As a B deficiency, there is less tolerance and will affect the quality calculation.
	<u>OTHER DEFICIENCIES – CATEGORY C DEFICIENCIES</u> These include a variety of other discrepancies that generally can be corrected. The contract/prescription must specify the maximum C deficiencies that are allowed, after which rework is required. In general, a greater number of these deficiencies can be allowed before it affects the overall results of the thinning activity, therefore there is some tolerance to a small number of these, particularly if there is a cost savings. Refer to contract for maximum Cat C deficiencies allowed. Recording the C deficiencies with a tally by type (i.e. C1, C2, etc.) is desirable.	
	C1	Tree was not completely severed from the stump.
	C2	Tree was hung up in another tree or other debris.
	C3	The stump was higher than the maximum allowed.
	C4	A “small tree” was left. Small trees are defined by the contract/prescription and reflect those trees that are required to be cut but do not pose a strong competition to the larger trees.
	C5	When girdling is required, tree should have been girdled or was girdled improperly. The prescription/contract must specify if this will be considered a C or B deficiency. As a C deficiency, there is tolerance for errors within the stand.
	C6	When pruning is required, tree should have been pruned or was pruned improperly. The prescription/contract must specify if this will be considered a C or B deficiency. As a C deficiency, there is more tolerance for errors within the stand.
NOTES		Evaluation Codes specific to other work required by the prescription/contract should be stated in this section, both deficiency and satisfactory codes. Additionally specify in this section if there are any deficiencies that will carry only a proportional value, for example “1/4 tree” for each pruning error. Thinning Inspectors should provide comments in this section relative to the overall stand condition, unusual or unique features or conditions observed, and other information that will be useful to the Government in evaluating the stand conditions.

Inspection Plot Form- Detailed Instructions

HEADER INFORMATION BLOCK	<u>EXPLANATION</u>
<u>STAND ID. GENERAL INFORMATION</u>	
a, b, c	Stand information (blocks a, b, c) is provided in the Silvicultural Summary Prescription.
d	Name of inspector when the Government is performing the inspection. This may be an inspection or a quality assurance check.
e	Date of inspection, regardless of who is conducting the inspection.
<u>CONTRACT INFORMATION</u>	
f, g, h	Name of Contractor, sub item number, and acres in unit
i	% of work accomplished: useful if this is a midway quality evaluation, however the unit quality is based on the completion of all plots in the unit or sub item. $\frac{\text{acres thinned to date}}{\text{unit/sub item acres}} \times 100 = \% \text{ work accomplished}$
j	Contract number for reference
k	Name of contractor's inspector when the contractor is conducting the inspection.
l	% of contract time left useful if this is a midway quality evaluation. $\frac{\text{Contract time used to date}}{\text{Total contract time}} \times 100 = \% \text{ contract time left.}$
CONTRACT SPECIFICATIONS- this is only a highlight of some of the requirements of the project; refer to the prescription and contract requirements for additional information.	
m	Acceptable stocking density in trees per acre provided in the prescription. An average, minimum, and maximum are provided.
n	Acceptable trees per plot are calculated from the acceptable trees per acre based on plot size. Calculate the minimum, maximum, and average. The minimum and maximum are required for evaluating trees on the plot. $\text{Trees /ac} \times \text{plot size} = \text{trees/plot}$ Example: 250 trees/ac x 1/50 th ac plot = 5 trees/plot
o	Spacing when spacing method is used for leave tree selection. Refer to prescription. Example: 10 x 10-foot average spacing.
p	Plot size is either specified in the contract or determined by inspector. The same plot size must be used on the entire unit; it should yield a minimum of 4-6 trees per plot (fully stocked plots). Utilize larger plot when variability in the stand density exists and it is desired to maintain that variability.
q	Minimum cut height is stated in the prescription/contract and reflects the shortest tree that is required to be cut, represented in inches of height from ground level.
r	Maximum cut d.b.h. is stated in the prescription and reflects the maximum d.b.h. for trees to be cut. Cutting trees larger is a violation.
s	Maximum Cat C is the maximum allowable Category C deficiencies allowed before rework is required. The maximum is stated in the prescription or contract.

<i>PLOT INSPECTION</i>	
HEADER INFORMATION BLOCK	<u>EXPLANATION</u>
Bottom of page	Thinning Quality calculation: ratio of Category A and B deficiencies to total satisfactory leave trees and credit for non-stocked points. $\text{Quality \%} = \frac{1.00 - \text{Total of col. 6 + col. 7 + col. 8 + col. 9}}{\text{Col 3} + \text{Col. 5}} \times 100$ Note: if there are a high number of deficiencies relative to allowable trees, the quality may be a negative number. This would indicate a need for re-work or other action to achieve acceptable work. Category C deficiency calculation after Cat C deficiencies are totaled on the unit $\text{Cat C trees per acre} = \frac{\text{Total col. 10}}{\text{No. of plots}} \times \text{reciprocal of plot size} = \text{tpa}$
Bottom of page	Contractor inspector must sign, date and certify that all information is correct and accurate before providing the records to the Government.
INSPECTION SHEET COLUMN	INSTRUCTIONS/EXPLANATIONS
1	Locate and mark plot center on the ground. Record the plot number in column 1.
2	- Locate and examine each tree on the plot selected for leave, and trees that have been cut. Record the Evaluation Code on the upper row of blocks for each tree on the plot, any tree that should have been left on the plot, and non-stocked points. Utilize the Tree Evaluation Codes on the back of the form. Do not include trees with a C deficiency (see column 10). - If required by the contract, record the species of each tree in the lower block. Record the species of a tree that was cut that should not have been with a diagonal line through it. - If required by the contract, estimate and record the average diameter and height of the trees after thinning the plot. These are estimates that can be used for the post-thinning stand condition. - In the remarks area, provide comments on tree condition, plot conditions, or other information useful to understand the inspection results
3	Based on the results of column 2, tally the satisfactory S1 and S2 leave trees on each plot and record in column 3. This number cannot exceed the maximum trees per plot (block n) and would only be less than the minimum when there are non-stocked points (N1, N2) or Category A deficiencies recorded on the plot. If this number is not within the max/min range, then the codes in column 2 need to be re-evaluated. When it exceeds the maximum, there are trees coded satisfactory that should be surplus (S3 or S4) or excess (Cat B). If it is below the minimum, and there are no points that satisfy the definition for non-stocked points (N1, N2), then an acceptable tree was cut (Cat A). The appropriate tree(s) should be re-coded. If an entire plot consists of trees which exceed the max. dbh, or in combination with Satisfactory (S1) trees, the maximum satisfactory leave trees in column 3 is the maximum trees/plot (block n).
4	Based on the results of column 2, tally the number of acceptable trees left because they exceeded the maximum d.b.h. (S3); or were trees that were ignored during the thinning (S4). These trees are surplus to those on the plot used in the acceptable trees per plot calculation based on 1) the prescription or 2) because these are in addition to the maximum acceptable trees per plot. This column is not used in the quality calculation, however, is useful in evaluating total trees remaining on the unit of all size classes.
5	Tally the non-stocked points on the plot (N1, N2). If the entire plot is an opening, the maximum number of N1 credit is the minimum acceptable trees/plot from block (n).
6	Tally the total of trees cut that should not have been cut (A1, A2).

COLUMN	INSTRUCTIONS/EXPLANATIONS
7	Tally the total trees that should have been left but were damaged during the operation (A3) or an error in selecting the correct leave tree (A4), hence the best or correct tree was cut.
8	Tally of excess trees which are trees that should have been cut (B1, B2) and cause there to be more trees per acre than acceptable, trees that are spaced too close to a satisfactory leave tree, or trees that should have been cut to meet contract specifications. Poor thinning quality can be improved by reworking areas with excess trees.
9	Tally of trees that should have been girdled or pruned or were done so incorrectly (B3 and B4). Poor quality can be improved by reworking areas with excess trees. Note that excess girdled or pruned trees <u>could</u> instead be a Category C deficiency. The FS must determine this prior to start work; specify on back of inspection form.
10	Tally other types of deficiencies on the plot. Tally by type of error utilizing the space below column 10. A maximum number of C deficiencies must be identified prior to start work (refer to prescription or contract). Areas that exceed this level must be re-worked. Refer to explanation for blocks. Category C deficiencies are not included in the quality calculation.
After all plots are taken, total the columns. Calculate thinning quality and determine if the Category C deficiencies are within the contract allowance. Thinning Quality (see block t) $\text{Quality \%} = 1.00 - \frac{\text{Totals of Column 6 + column 7 + Column 8 + Column 9}}{\text{Column 3 + Column 5}} \times 100$ <i>Note: if the numerator exceeds the denominator, a negative thinning quality will result.</i> Category C trees per acre = $\frac{\text{Total of Column 10}}{\text{No. of plots}} \times \text{reciprocal of plot size}$ <i>Compare results with blocks.</i>	
Additional calculations that may be conducted by the Government to determine if thinning met the prescription goals. <u>Average number of satisfactory leave trees (S1 and S2 only) per plot:</u> $\frac{\text{Sum of Column 3}}{\text{No. of plots}} = \text{avg. trees /plot}$ Average number of satisfactory trees (S1, S2 only) per acre: $\frac{\text{Sum of Column 3}}{\text{No. of plots}} \times \text{reciprocal of plot size} = \text{trees per acre}$ Average trees per acre in the diameter range being thinned (S1, S2, A3, A4, B1, B2 trees): $\frac{[\text{Sum of Columns 3 + 7+8}]}{\text{No. of plots}} \times \text{reciprocal of plot size} = \text{trees per acre}$ <u>Average trees per plot in the diameter range being thinned:</u> $\frac{[\text{Sum of Columns 3 +7+8}]}{\text{No. of plots}} = \text{avg. trees /plot}$ Average Total trees per acre (all trees larger than “small trees” and min. cut tree height): $\frac{[\text{Sum of Columns 3 + 4 + 6 + 7 + 8.}]}{\text{No. of plots}} \times \text{reciprocal of plot size} = \text{trees per acre}$	